

JOÃO FONSECA

COMPUTER ENGINEER



PERFIL

Computer Engineer, with a Bachelor's degree from Universidade Lusófona, currently completing a Master's degree in Computer Engineering, specializing in Data Engineering, at ISEP. Throughout my academic background, I have developed skills in programming, databases, and software development. I am looking for an opportunity to apply my knowledge, continue learning, and develop new skills, while remaining open to different challenges and areas within the field of IT.

PROJECTS

Analysis and Application of Machine Learning Techniques to Musical Data

Objective: To analyze, model, and compare Machine Learning algorithms applied to music popularity prediction and genre classification.

Technical Value: Incorporates regression, classification, cross-validation, and regularization techniques, demonstrating best practices in applied Machine Learning.

Polyglot Persistence – Urban Transportation System

Objective: To design and implement a modular architecture (React and Node.js) for a public transportation system, focused on scalability, performance, and polyglot persistence.

Technical Value: Optimizes data storage by selecting the most appropriate technology for each use case: PostgreSQL for critical data, MongoDB for flexible data, Redis for caching, and Neo4j for route management.

Football Scouting RAG System

Objective: Develop a football scouting system based on RAG, capable of searching for players using natural language and generating structured reports.

Technical Value: Compares two RAG architectures using ChromaDB, Ollama, scikit-learn, and Flask, integrating hybrid retrieval, embeddings, text mining, post-retrieval validation, and hallucination reduction techniques.

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🚗 Carta de condução: B

ACADEMIC BACKGROUND

2025 - PRESENT

INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

- Master's Degree in Computer Engineering – Data Engineering

2021 - 2024

UNIVERSIDADE LUSÓFONA DO PORTO

- Bachelor's Degree in Computer Engineering

SKILLS

Programming Languages and Modelling: Python, Java, C, C++, JavaScript, Html e CSS, Dart;

Databases: MySQL, PostgreSQL, Oracle, Mongo, Redis, Neo4j, Firebase

Tools and Frameworks: Git, Docker, React, React Native, Flutter, Node.js, Next.js, Express.js

LANGUAGES

- Portuguese | Native
- English | Intermediate